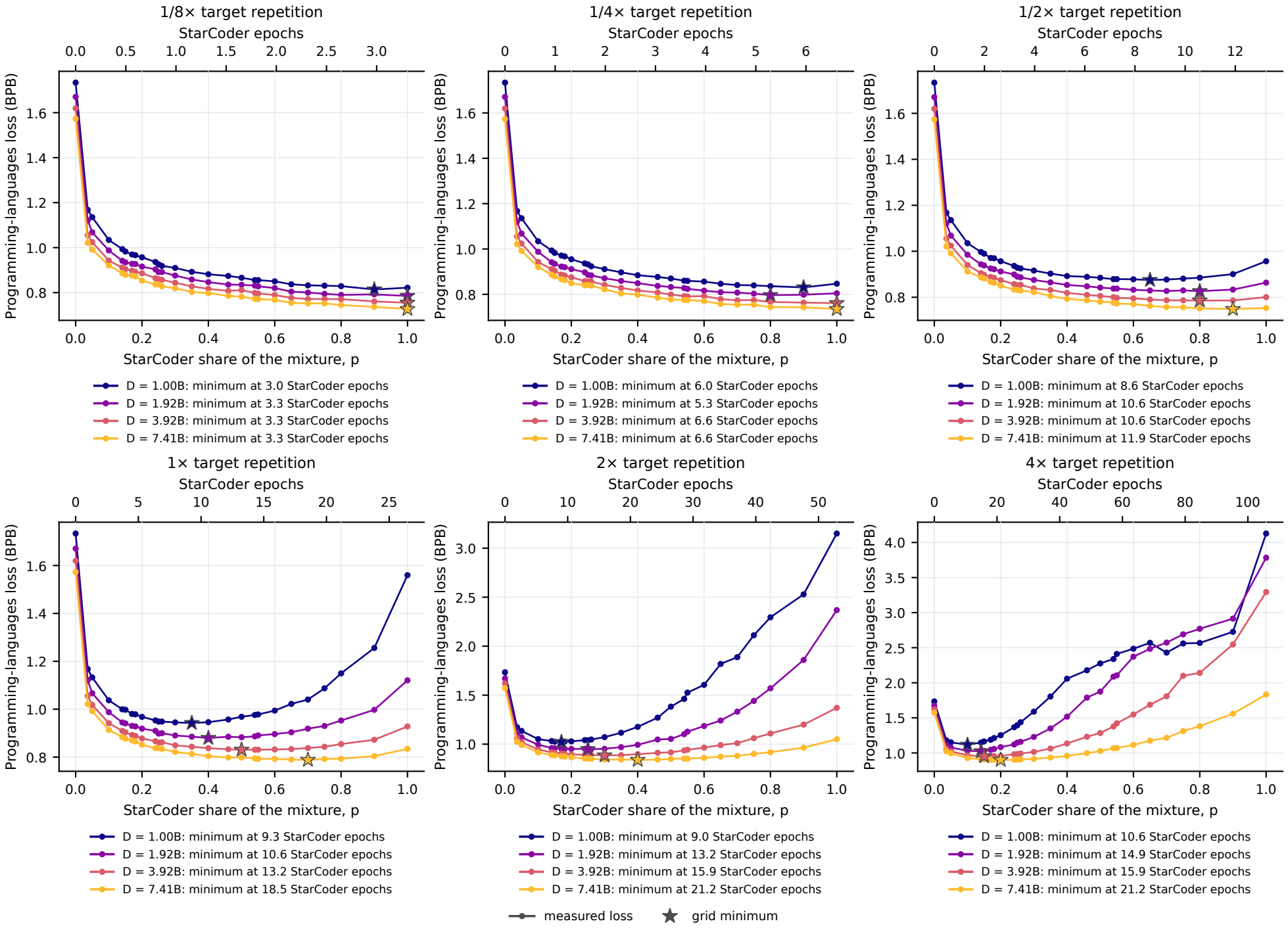
1/8× target repetition
StarCoder epochs
0.0 0.5 1.0 1.5 2.0 2.5 3.0
Programming-languages loss (BPB)
1.6 1.4 1.2 1.0 0.8
0.0 0.2 0.4 0.6 0.8 1.0
StarCoder share of the mixture, p
D = 1.00B: minimum at 3.0 StarCoder epochs
D = 1.92B: minimum at 3.3 StarCoder epochs
D = 3.92B: minimum at 3.3 StarCoder epochs
D = 7.41B: minimum at 3.3 StarCoder epochs
1/4× target repetition
StarCoder epochs
0 1 2 3 4 5 6
Programming-languages loss (BPB)
1.6 1.4 1.2 1.0 0.8
0.0 0.2 0.4 0.6 0.8 1.0
StarCoder share of the mixture, p
D = 1.00B: minimum at 6.0 StarCoder epochs
D = 1.92B: minimum at 5.3 StarCoder epochs
D = 3.92B: minimum at 6.6 StarCoder epochs
D = 7.41B: minimum at 6.6 StarCoder epochs
1/2× target repetition
StarCoder epochs
0 2 4 6 8 10 12
Programming-languages loss (BPB)
1.6 1.4 1.2 1.0 0.8
0.0 0.2 0.4 0.6 0.8 1.0
StarCoder share of the mixture, p
D = 1.00B: minimum at 8.6 StarCoder epochs
D = 1.92B: minimum at 10.6 StarCoder epochs
D = 3.92B: minimum at 10.6 StarCoder epochs
D = 7.41B: minimum at 11.9 StarCoder epochs
1× target repetition
StarCoder epochs
0 5 10 15 20 25
Programming-languages loss (BPB)
1.6 1.4 1.2 1.0 0.8
0.0 0.2 0.4 0.6 0.8 1.0
StarCoder share of the mixture, p
D = 1.00B: minimum at 9.3 StarCoder epochs
D = 1.92B: minimum at 10.6 StarCoder epochs
D = 3.92B: minimum at 13.2 StarCoder epochs
D = 7.41B: minimum at 18.5 StarCoder epochs
2× target repetition
StarCoder epochs
0 10 20 30 40 50
Programming-languages loss (BPB)
3.0 2.5 2.0 1.5 1.0
0.0 0.2 0.4 0.6 0.8 1.0
StarCoder share of the mixture, p
D = 1.00B: minimum at 9.0 StarCoder epochs
D = 1.92B: minimum at 13.2 StarCoder epochs
D = 3.92B: minimum at 15.9 StarCoder epochs
D = 7.41B: minimum at 21.2 StarCoder epochs
4× target repetition
StarCoder epochs
0 20 40 60 80 100
Programming-languages loss (BPB)
4.0 3.5 3.0 2.5 2.0 1.5 1.0
0.0 0.2 0.4 0.6 0.8 1.0
StarCoder share of the mixture, p
D = 1.00B: minimum at 10.6 StarCoder epochs
D = 1.92B: minimum at 14.9 StarCoder epochs
D = 3.92B: minimum at 15.9 StarCoder epochs
D = 7.41B: minimum at 21.2 StarCoder epochs
measured loss
grid minimum